

Nagabhushan S N

Contact Information

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Research Interests

- ◇ 3D Computer Vision
- ◇ Computer Graphics
- ◇ Image and Video Signal Processing
- ◇ Machine Learning

Education

Ph.D. - Indian Institute of Science (IISc), Bengaluru Aug 2018 - May 2024

Dept of Electrical Communication Engineering

- ◇ Advisor : Dr. Rajiv Soundararajan
- ◇ Thesis Reviewers : Dr. Aniket Bera (Purdue University), Dr. Kaushik Mitra (IIT-Madras)
- ◇ CGPA : 9.59
- ◇ Recipient of Prime Minister's Research Fellowship (PMRF)

B.E. - PES Institute of Technology, Bengaluru Aug 2012 - May 2016

Dept of Electronics and Communication Engineering

- ◇ Project Advisors : Dr. Vamsi Krishna, Dr. Sanjeev Gurugopinath
- ◇ CGPA : 9.93 (Gold Medallist)
- ◇ GATE Score : 810 (AIR 233)

P.U.C. - Govt [Ex-MPL] College, Ballari Jun 2010 - May 2012

- ◇ Specialization : Science (PCME)
- ◇ Class XII Score : 96% (District Topper)
- ◇ K-CET Rank : 207

School - Bala Bharathi, Ballari Jun 1998 - Mar 2010

- ◇ Class X Score : 96.8% (District Topper)

Course Work

Mathematics : Linear Algebra, Probability and Random Process, Linear and Non-Linear Optimization
Electrical : Computer Vision, Digital Image Processing, Digital Video: Perception and Algorithms, Machine Learning for Signal Processing, Detection and Estimation Theory

Certifications

- ◇ **Algorithmic Toolbox** by University of California San Diego on Coursera in Oct 2016.

Computer Skills

Programming Languages : Python (+ PyTorch), Java, Android, MATLAB
Platforms : Windows, LINUX, MacOS
Softwares : L^AT_EX
AI tools : GitHub Copilot, Claude Code, Codex

Industrial Experience

Aina Tech Aug 2026 - Present

- ◇ Role : Research Scientist
- ◇ Sparse Viewpoint 4D Reconstruction: 4D reconstruction models based on 3DGS require hundreds of viewpoints/cameras for photo-realistic reconstruction. We are working on reducing the number of viewpoints required for 4D reconstruction by using video diffusion models to bootstrap the 4D reconstruction by generating additional high quality viewpoints.

SpreeAI Corporation May 2024 - June 2026

- ◇ Role : Research Scientist
- ◇ Team : Long Term Research
- ◇ Garment Fit and Size: Developed the first model to predict size accurate 3D representations of garments from a single image (calibrated flat-garment image). The work involved designing a synthetic flat garment generation pipeline that creates realistic flat-garments. The garments are represented using sewing patterns which can be draped on personalized human body models (SMPL) to visualize the size and fit.

Cisco Systems India Pvt Ltd Aug 2016 - Jul 2018

- ◇ Role : Software Engineer
- ◇ Team : Software Integration and Orchestration (Advanced Services)
- ◇ Developed modules on top of Cisco Network Service Orchestrator (NSO) to automate configuring customer's network devices.
- ◇ Developed an automation tool to generate Java POJO classes based on network architecture defined in yang files.

Elseem Inc. Jun 2015 - Jul 2015

- ◇ Role : Intern
- ◇ Team : Research and Development
- ◇ Vehicle Tracking System: Developed software for Particle device to interface with GPS module and update location to firebase server. Also developed an android app for the users to track their vehicle using their smartphone.

Academic Experience

PES University (PESU) Jan 2021 - May 2023

- ◇ Visiting Faculty.
- ◇ Mathematics for Machine Learning: Jan 2022 semester, Jan 2023 semester.
- ◇ Matrix Theory: Jan 2021 semester.

Indian Institute of Science (IISc) Oct 2020 - May 2022

- ◇ Teaching Assistant.
- ◇ Courses: Matrix Theory, Digital Image Processing.

Publications

- ◇ Conferences:

- Gurutva Patle, Nilay Girgaonkar, Nagabhushan Somraj, and Rajiv Soundararajan. AD-GS: Alternating densification for sparse-input 3d gaussian splatting. In *Proceedings of the ACM SIGGRAPH Asia*, December 2025.
- Kapil Choudhary, Nagabhushan Somraj, and Rajiv Soundararajan. Bidirectional flow fields for sparse input novel view synthesis of dynamic scenes. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2025.
- Nagabhushan Somraj, Kapil Choudhary, Sai Harsha Mupparaju, and Rajiv Soundararajan. Factorized motion fields for fast sparse input dynamic view synthesis. In *Proceedings of the ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)*, 2024.
- Nagabhushan Somraj, Adithyan Karanayil, and Rajiv Soundararajan. SimpleNeRF: Regularizing sparse input neural radiance fields with simpler solutions. In *Proceedings of the ACM SIGGRAPH Asia*, December 2023.
- Nagabhushan Somraj and Rajiv Soundararajan. ViP-NeRF: Visibility prior for sparse input neural radiance fields. In *Proceedings of the ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)*, 2023.
- Nagabhushan Somraj, Pranali Sancheti, and Rajiv Soundararajan. Temporal view synthesis of dynamic scenes through 3d object motion estimation with multi-plane images. In *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2022.
- Vijayalakshmi Kanchana, Nagabhushan Somraj, Suraj Yadwad, and Rajiv Soundararajan. Revealing disocclusions in temporal view synthesis through infilling vector prediction. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- ◇ Journals:
 - Nagabhushan Somraj, Sai Harsha Mupparaju, Adithyan Karanayil, and Rajiv Soundararajan. Simple-RF: Regularizing sparse input radiance fields with simpler solutions. *ACM Transactions on Graphics (TOG)*, 45(3):1–24, 2026.
 - Nagabhushan Somraj, Manoj Surya Kashi, S. P. Arun, and Rajiv Soundararajan. Understanding the perceived quality of video predictions. *Signal Processing: Image Communication (SPIC)*, 102:116626, 2022.

Peer Reviews

- ◇ IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR): 2026, 2025, 2024
- ◇ IEEE International Conference on Computer Vision (ICCV): 2025, 2023
- ◇ European Conference on Computer Vision (ECCV): 2026, 2024 (**Outstanding Reviewer**)
- ◇ IEEE Winter Conference on Applications of Computer Vision (WACV): 2025, 2024 (**Outstanding Reviewer**), 2023
- ◇ AAAI Conference on Artificial Intelligence: 2026, 2025
- ◇ British Machine Vision Conference (BMVC): 2024, 2023, 2022
- ◇ Asian Conference on Computer Vision (ACCV): 2024
- ◇ Pacific Graphics: 2024
- ◇ IEEE International Symposium on Mixed and Augmented Reality (ISMAR): 2025, 2024, 2023, 2022 (**Highly useful review**)
- ◇ IEEE Virtual Reality (VR): 2024, 2023
- ◇ IEEE Transactions on Image Processing (TIP): 2025
- ◇ IEEE Transactions on Circuit Systems and Video Technologies (TCSVT): 2024

- ◇ IEEE Transactions on Computational Imaging (TCI): 2024
- ◇ Journal of Electronic Imaging (JEI): 2022, 2021, 2020
- ◇ ACM Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP): 2025
- ◇ International Conference on Signal Processing and Communications (SPCOM): 2024
- ◇ IEEE Global Communications Conference (GLOBECOM): 2024

Invited Talks

- ◇ Sparse Input View Synthesis with Neural Radiance Fields (NeRF)
 - EECS Symposium 2024 (Best Presentation Award).
- ◇ NeRF: Neural Radiance Fields
 - At GE Health Care India, May 2025.
 - At Qualcomm India Pvt Ltd, Feb 2023.
 - Guest Lecture for the course, Deep Learning for Computer Vision, 2022.
- ◇ Temporal View Synthesis of Dynamic Scenes
 - Student Research Seminar Series, IISc, Feb 2022.
- ◇ Naturalness Assessment for Video Prediction
 - Network Seminar Series, IISc, Oct 2020.
- ◇ Generative Adversarial Networks (GANs) and their Applications
 - PES University, Jan 2020.

Projects

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| Face Detection using Aggregate Channel Features <ul style="list-style-type: none"> ◇ Digital Image Processing course project. ◇ Implemented in Python. ◇ Based on paper: | 2019 |
| Intra-Campus Wi-Fi Calling System <ul style="list-style-type: none"> ◇ B.E. Final Year Project. ◇ Developed a system to make voice calls and transmit messages within a campus using Wi-Fi and without internet. Designed a simple protocol to handle messaging and calls. TCP is used to transmit packets. Applied Voice Activity Detection to suppress non-voice packets thereby reducing delay. Also developed a server to connect calls and an android app for client-side. | 2016 |
| Android Apps <ul style="list-style-type: none"> ◇ Finance Manager: Helps managing the finances of individuals, with added features such as statistics, automatic entry of transactions from bank messages, export to PDF and backup/restore. This is developed in Java using Android Studio. ◇ One Touch Settings: Adds a button panel in notification drawer to control frequently used settings. ◇ Google Play Developer Profile: Link. | 2014–17 |

Honors & Awards

- ◇ Finalist in IKDD Doctoral Dissertation in Data Science Award 2025.
- ◇ Recipient of Best Presentation Award at EECS Symposium 2024.
- ◇ Recipient of Prime Minister's Research Fellowship (PMRF): 2020, 2021, 2022.

- ◇ Recipient of 'You Amaze One' award in Cisco as recognition from Raghuveer Krishna, Manager: 2017.
- ◇ Recipient of MHRD scholarship (Govt of India): 2012, 2013, 2014, 2015.
- ◇ Recipient of MRD scholarship: 2014 Dec, 2015 May, 2015 Dec, 2016 May.
- ◇ 1st Place in Circuithon competition conducted by IEEE PES University Student Branch.
- ◇ 2nd Prize in Technical Writing Competition organized as a part of LitFest 2014 in PESIT.
- ◇ Volunteered and coordinated the Science Exhibition 'Prakalpa' held in PESIT (2014).